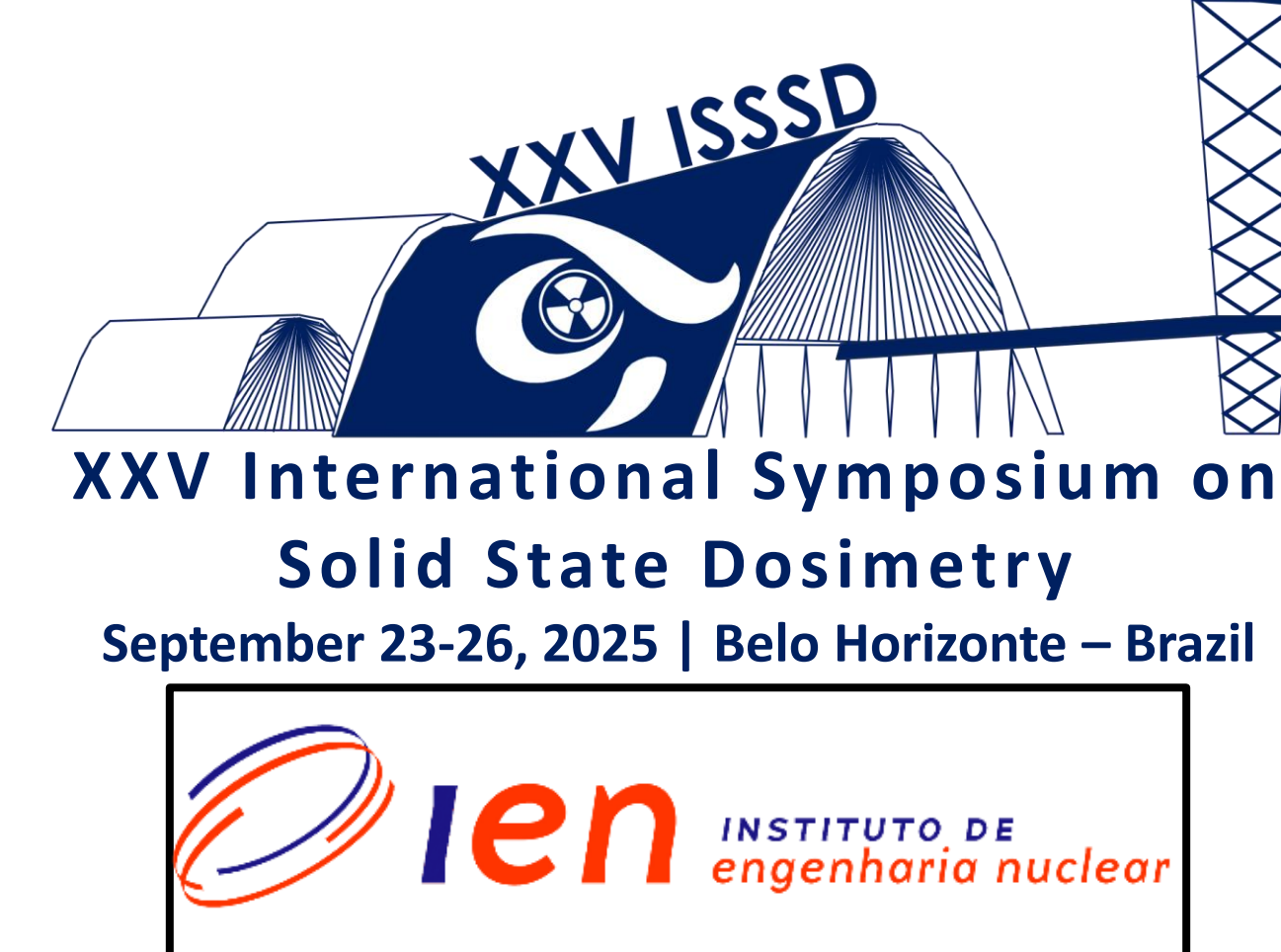


Determination of the mass attenuation coefficient of polymers using Monte Carlo N-Particle simulation



J.P.C. Andrade^{23*}, R.S.F Dam¹, C.M. Salgado¹, W.L. Salgado¹

¹ Instituto de Engenharia Nuclear (IEN), Divisão do Radiofármacos, Cidade Universitária, RJ, Brazil

² Instituto de Engenharia Nuclear (IEN), Divisão do ensino, Cidade Universitária, RJ, Brazil

³ Universidade Federal do Rio de Janeiro (UFRJ), Cidade Universitária Ilha do Fundão, RJ, Brazil

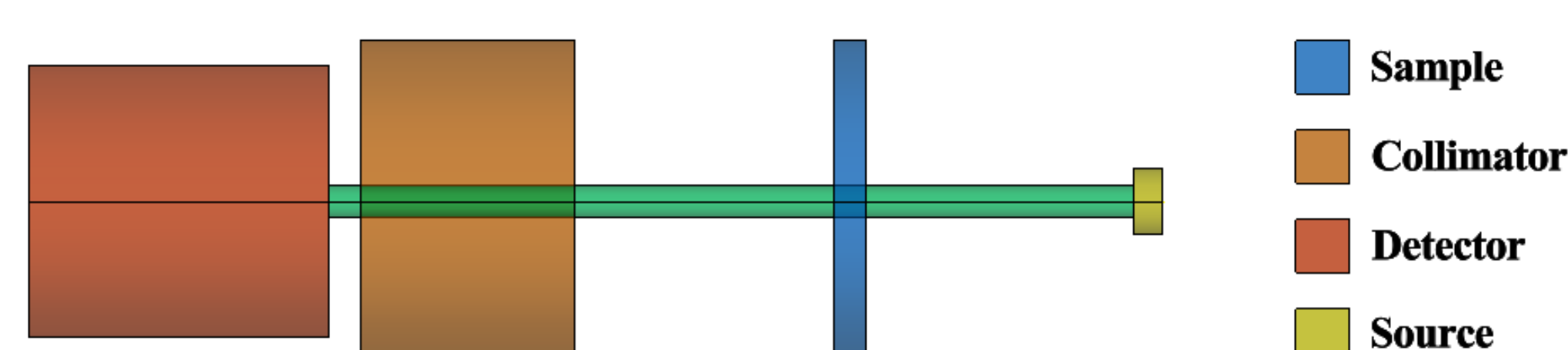
*jp.vc9999@gmail.com

1. Introduction

This study evaluates the potential of 3D-printed polymers acrylonitrile butadiene styrene (ABS), polyethylene terephthalate glycol (PETG), and polylactic acid (PLA), for gamma radiation mass attenuation coefficient. Using the MCNPX Monte Carlo code, the mass attenuation coefficients (MAC) for these materials were calculated. The computational model simulated the interaction of photons from standard sources ²⁴¹Am, ¹³⁷Cs, and ⁶⁰Co with the samples under a narrow-beam geometry, using a 2"x2" NaI(Tl) detector to count the number of interactions. The methodology was validated by comparing the results for an aluminum control sample with reference data from the NIST XCOM [1] database, which showed good agreement. This research provides fundamental data on the attenuation capabilities of these polymers, aiming for their use in new, low-cost shielding and support for detectors in the nuclear field.

2. Materials and Methods

The simulation modeled a narrow-beam transmission experiment. A monoenergetic pencil beam of photons was directed perpendicularly onto a circular polymer sample (6.0 cm diameter, 0.6 cm thickness). A simulated 2"x2" NaI(Tl) scintillation detector was placed 15 cm from the source to measure the transmitted radiation. Two geometries were simulated: one with a lead collimator placed before the detector and one without.



To calculate the mass attenuation coefficient (μ/ρ) was used the Beer-Lambert law, based on the incident (I_0) and transmitted (I) photon intensities recorded by the detector, x the thickness of the sample, and ρ the density.

$$I = I_0 e^{-(\mu/\rho) \cdot (\rho x)}$$

(1)

The simulated photon sources were ²⁴¹Am (59.54 keV), ¹³⁷Cs (661.66 keV), and ⁶⁰Co (1173.23 and 1332.49 keV). The number of photons reaching the detector was recorded using the F8 pulse height tally. To ensure high statistical precision, simulations were run for 100 million particle histories, resulting in a relative error below 1%.

The elemental compositions and densities of the polymers, in the table 1, were used in the MCNPX input. The compositions for PLA (C₃H₄O₂) and PETG (C₁₄H₂₀O₅S) were calculated from their chemical formulas. For ABS, a more complex copolymer, the composition and density were obtained from the study by Moayyad Alssabbagh [2]. For validation, the values obtained of MAC were compared with the value of NIST

Polymer	C (%)	H (%)	O (%)	S (%)	N (%)	Density (g/cm ³)
PLA	50.0025	5.5942	44.4032	-	-	1.29
PETG	55.9821	6.7106	26.6321	10.0675	-	1.30
ABS	85.51	5.26	1.65	0.08	5.26	1.09

Table 1: Elemental composition and densities of simulated polymers.

3. Results

Table 2: Mass attenuation coefficient from the simulations.

Energy (keV)	Mass attenuation coefficient (μ/ρ) (cm ² /g)							
	Without collimator				With collimator			
	ABS	PLA	PETG	Al	ABS	PLA	PETG	Al
59,6	1,82E-01	1,84E-01	2,07E-01	2,62E-01	1,87E-01	1,91E-01	2,15E-01	2,80E-01
662	8,23E-02	8,09E-02	8,17E-02	7,39E-02	8,26E-02	8,13E-02	8,21E-02	7,44E-02
1173,2	6,23E-02	6,13E-02	6,19E-02	5,60E-02	6,28E-02	6,17E-02	6,23E-02	5,65E-02
1332,5	5,85E-02	5,75E-02	5,81E-02	5,27E-02	5,88E-02	5,78E-02	5,84E-02	5,29E-02

Table 3: Reference mass attenuation coefficients from NIST.

Energy (keV)	Mass attenuation coefficient (μ/ρ) (cm ² /g)			
	ABS	PLA	PETG	AL
59.6	1.88E-01	1.91E-01	2.15E-01	2.80E-01
662	8.29E-02	8.15E-02	8.22E-02	7.47E-02
1173.2	6.31E-02	6.21E-02	6.27E-02	5.68E-02
1332.5	5.91E-02	5.81E-02	5.87E-02	5.22E-02

Table 4: Relative error (%) between simulation and NIST values.

Energy (keV)	Relative error (%)							
	Without collimator				With collimator			
	ABS	PLA	PETG	Al	ABS	PLA	PETG	Al
59.6	3.18	3.77	3.88	6.44	0.20	0.21	0.15	0.04
662	0.72	0.68	1.27	0.99	0.29	0.24	0.08	0.36
1173.2	1.28	1.27	1.25	1.26	0.51	0.51	0.48	0.47
1332.5	1.01	0.99	0.99	1.03	0.49	0.49	0.47	0.49

4. Conclusions

This study validated the MCNPX simulation as a highly accurate method for determining the mass attenuation coefficients of 3D-printed polymers. The inclusion of a collimator proved critical, drastically reducing the relative error from a high of 6.44% (observed with the low-energy ²⁴¹Am source) to consistently below 0.52% for all materials in the final setup. This significant improvement, especially at low energies, highlights the successful mitigation of scattering effects and confirms the method's robustness. The findings support the use of these polymers in nuclear applications,

Acknowledgements

The authors gratefully acknowledge scholarships from Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq).

References

- [1] ALSSABBAGH, Moayyad et al. Evaluation of 3D printing materials for fabrication of a novel multifunctional 3D thyroid phantom for medical dosimetry and image quality. Radiation Physics and Chemistry, v. 135, p. 106–112, 2017.
- [2] NIST National Institute of Standards and Technology. Available online. <https://physics.nist.gov/PhysRefData/Xcom/html/xcom1.html>.